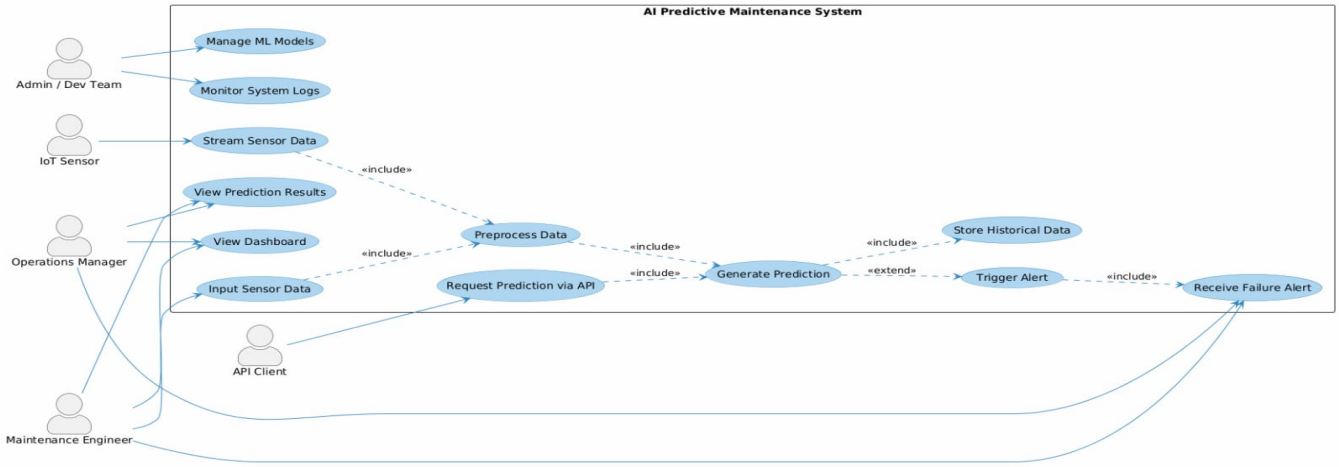
AI Predictive Maintenance System

Analysis Models



1. Actor Profiles

Actor	Actor Profile	Role Description
Maintenance Engineer	A technical user responsible for monitoring equipment condition and performing maintenance activities	Receives alerts, reviews predictions, and takes action to prevent equipment failure
System Operator	A user responsible for operating the system and ensuring smooth data flow	Inputs sensor data, monitors system performance, and interacts with the API
Admin	A system administrator responsible for managing system operations and monitoring performance	Manages users, monitors system behavior, and ensures system reliability
Predictive Maintenance Engine (Virtual Actor)	An AI-based system component that analyzes sensor data and predicts failures	Processes data, extracts features, generates predictions, and triggers alerts

2. Actors and Goals

Actor(s)	Goal(s)	Privilege Level
Maintenance Engineer	View predictions, receive alerts, take maintenance actions	User
System Operator	Input sensor data, access API, monitor system outputs	User
Admin	Manage system, monitor performance, ensure system stability	Admin
Predictive Maintenance Engine	Analyze sensor data, detect patterns, predict failures, generate alerts	System

3. Summarized Use Cases

Use Case	Brief Description
Collect Sensor Data	The system collects sensor readings such as temperature, vibration, and pressure from equipment
Preprocess Data	The system cleans and prepares raw sensor data for analysis
Feature Engineering	The system extracts meaningful features from processed data
Predict Equipment Failure	The system analyzes data using AI models to predict failure likelihood
Generate Alert	The system generates alerts when a high risk of failure is detected
View Prediction Results	The user views equipment status and prediction outputs
Access API	The user sends data to the system and receives predictions through an API

4. Semi-Detailed Use Cases

Use Case 1: Collect Sensor Data

Field	Collect Sensor Data
Primary Actor	System Operator / Sensors
Main Successful Scenario	1. Sensors generate data 2. Data is sent to system 3. System stores data
Alternative Scenario	Sensor failure → "Sensor error" Missing data → "Incomplete sensor data"

Use Case 2: Preprocess Data

Field	Preprocess Data
Primary Actor	ML System
Main Successful Scenario	1. Load raw data 2. Handle missing values 3. Remove noise 4. Normalize data
Alternative Scenario	Invalid data → discard or log error

Use Case 3: Feature Engineering

Field	Feature Engineering
Primary Actor	ML System
Main Successful Scenario	1. Calculate moving averages 2. Compute rate of change 3. Generate statistical features
Alternative Scenario	Feature error → log issue

Use Case 4: Predict Equipment Failure

Field	Predict Equipment Failure
Primary Actor	ML System
Main Successful Scenario	1. Input features into model 2. Model analyzes data 3. Generate prediction
Alternative Scenario	Model error → "Prediction failed"

Use Case 5: Generate Alert

Field	Generate Alert
Primary Actor	ML System
Main Successful Scenario	1. Check prediction 2. If high risk → generate alert 3. Send alert to user
Alternative Scenario	No risk → no alert

Use Case 6: View Prediction Results

Field	View Prediction Results
Primary Actor	Maintenance Engineer
Main Successful Scenario	1. User logs in 2. Opens dashboard 3. Views results
Alternative Scenario	No data → "No results found"

Use Case 7: Access API

Field	Access API
Primary Actor	System Operator
Main Successful Scenario	1. Send API request 2. System processes data 3. Return prediction response
Alternative Scenario	API failure → return error response

5. Full Detailed Use Cases

Use Case 1: Collect Sensor Data

Field	Description
Use Case Name	Collect Sensor Data
Primary Actor	System Operator / Sensors
Description (Goal)	Collect real-time sensor data from industrial equipment
Precondition	Sensors are active and system is running
Basic Path (Scenario)	1. Sensors generate data (temperature, vibration, pressure) 2. Data is transmitted to system 3. System validates input 4. Data is stored in database
Post Condition	Sensor data is stored and ready for processing
Alternative Paths	Missing data → "Incomplete sensor data" Sensor failure → log error

Use Case 2: Preprocess Data

Field	Description
Use Case Name	Preprocess Data
Primary Actor	ML System
Description (Goal)	Clean and prepare raw sensor data for analysis
Precondition	Raw sensor data is available
Basic Path (Scenario)	1. Load raw data 2. Handle missing values 3. Remove noise/outliers 4. Normalize data
Post Condition	Cleaned dataset ready for feature engineering
Alternative Paths	Invalid data → discard or flag Corrupted data → log error

Use Case 3: Feature Engineering

Field	Description
Use Case Name	Feature Engineering
Primary Actor	ML System
Description (Goal)	Extract meaningful features from sensor data
Precondition	Clean data is available
Basic Path (Scenario)	1. Compute moving averages 2. Calculate rate of change 3. Generate statistical features (mean, std, variance)
Post Condition	Feature set ready for ML model
Alternative Paths	Feature calculation error → log failure

Use Case 4: Predict Equipment Failure

Field	Description
Use Case Name	Predict Equipment Failure
Primary Actor	ML System
Description (Goal)	Predict whether equipment will fail or not
Precondition	Feature dataset is available
Basic Path (Scenario)	1. Input features into ML model 2. Model processes data (XGBoost / LSTM) 3. Generate prediction (Normal / Failure Risk)
Post Condition	Prediction result stored in system
Alternative Paths	Model error → return failure message Invalid input → reject request

Use Case 5: Generate Alert

Field	Description
Use Case Name	Generate Alert
Primary Actor	ML System
Description (Goal)	Notify users when failure risk is detected
Precondition	Prediction result is available
Basic Path (Scenario)	1. Check prediction result 2. If risk is high → trigger alert 3. Send alert to Maintenance Engineer
Post Condition	Alert successfully delivered
Alternative Paths	No risk detected → no alert generated

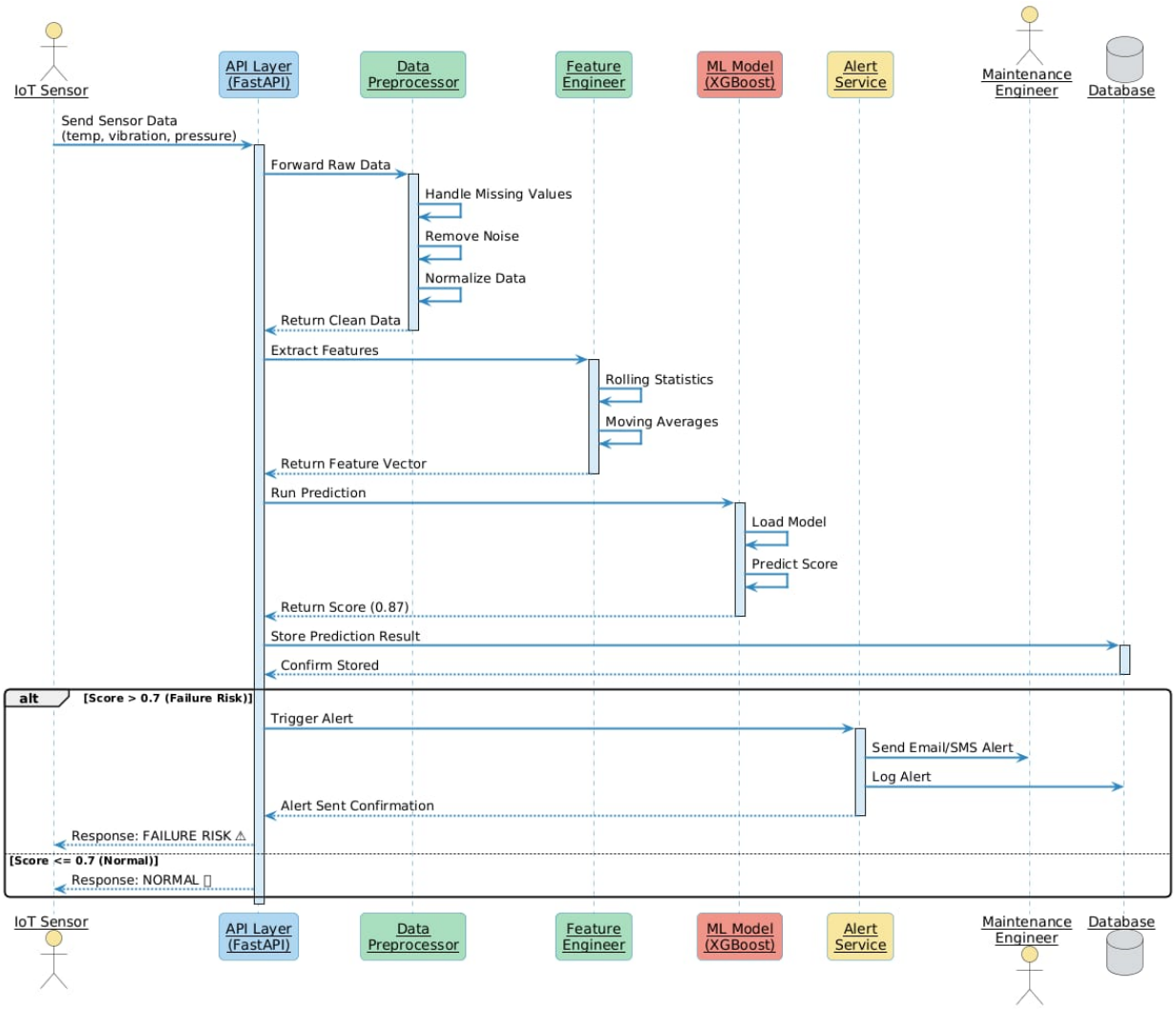
Use Case 6: View Prediction Results

Field	Description
Use Case Name	View Prediction Results
Primary Actor	Maintenance Engineer
Description (Goal)	View equipment status and AI predictions
Precondition	System has generated prediction
Basic Path (Scenario)	1. User logs into system 2. Opens dashboard 3. Views machine status and failure probability
Post Condition	User understands system output
Alternative Paths	No data → "No results available"

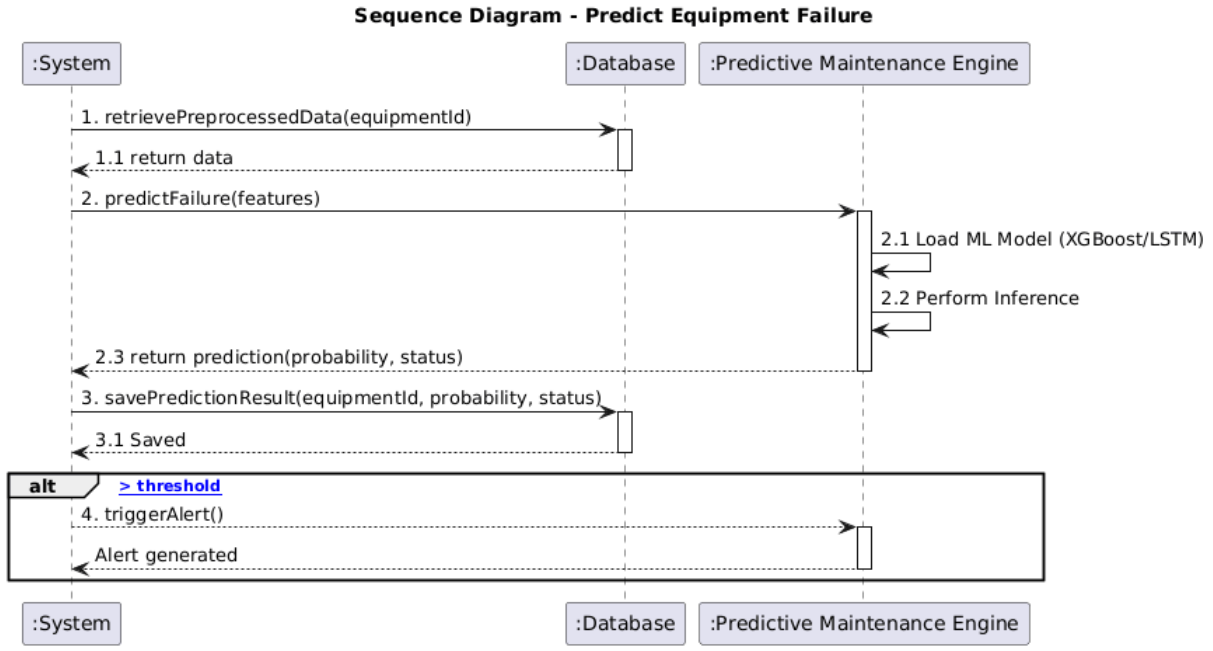
Use Case 7: Access API

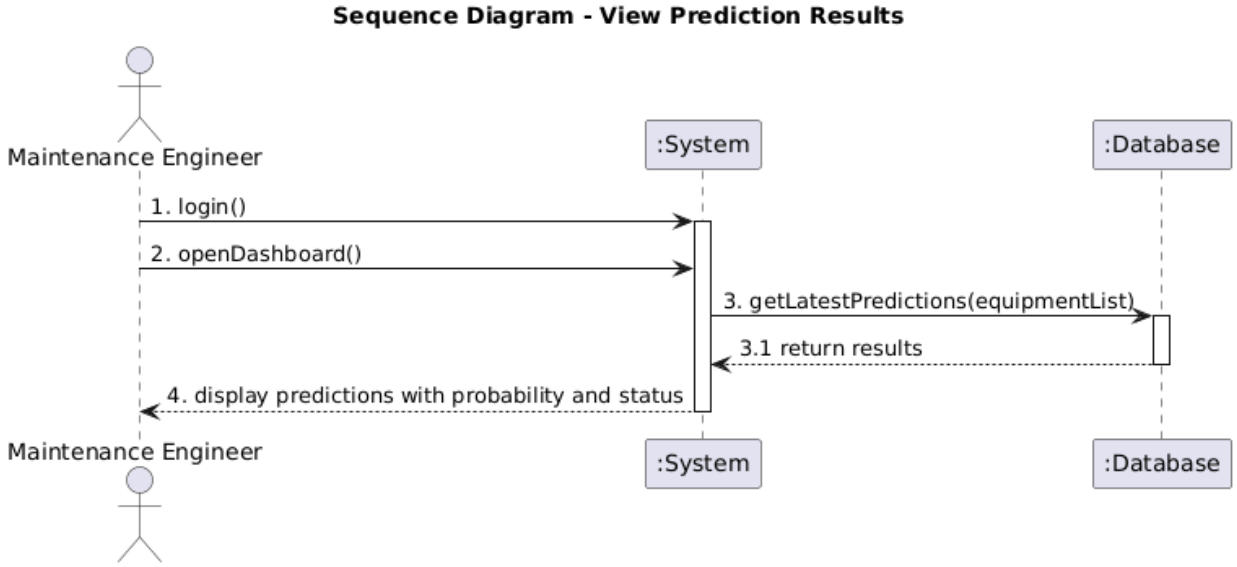
Field	Description
Use Case Name	Access API
Primary Actor	System Operator
Description (Goal)	Allow external systems/users to send data and receive predictions
Precondition	API service is running
Basic Path (Scenario)	1. Send request with sensor data 2. System processes request 3. ML model generates prediction 4. Response returned via API
Post Condition	Prediction response delivered successfully
Alternative Paths	API failure → return error response Invalid request → reject input

4 *System Sequence Diagram:*

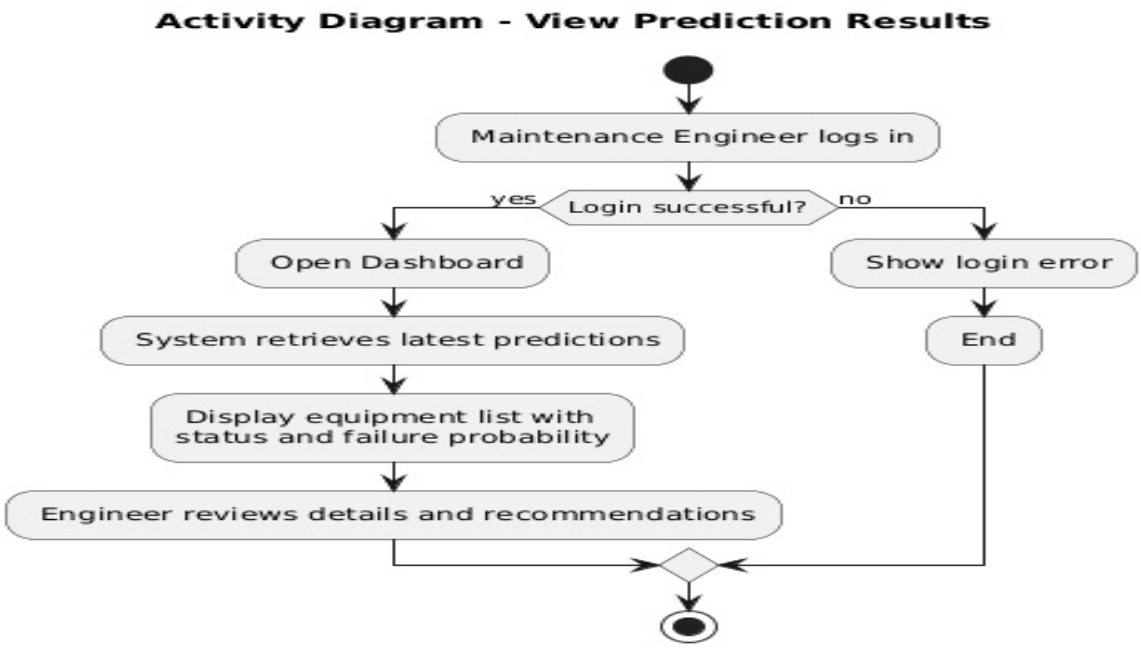


5. Sequence Diagram:

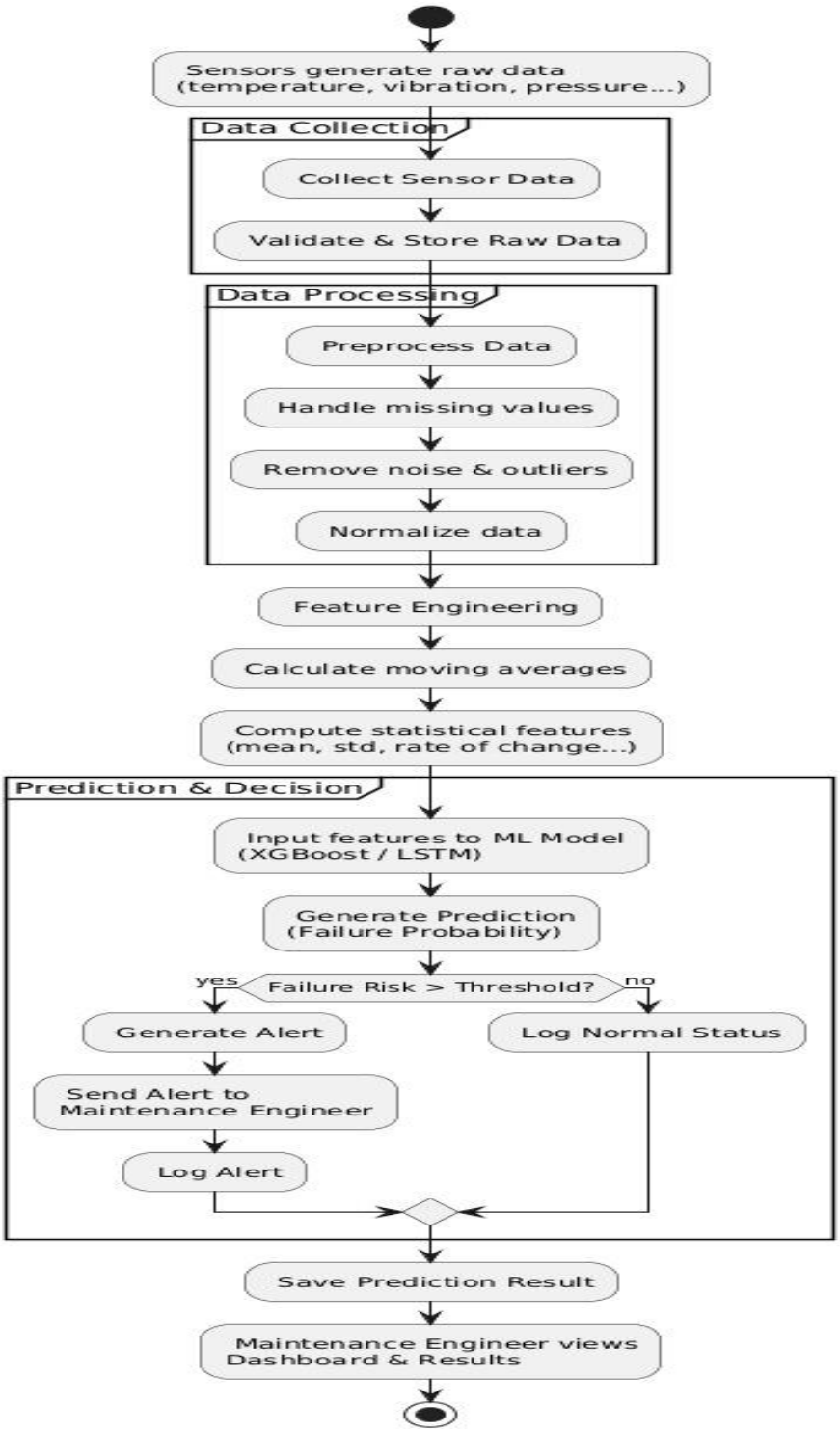




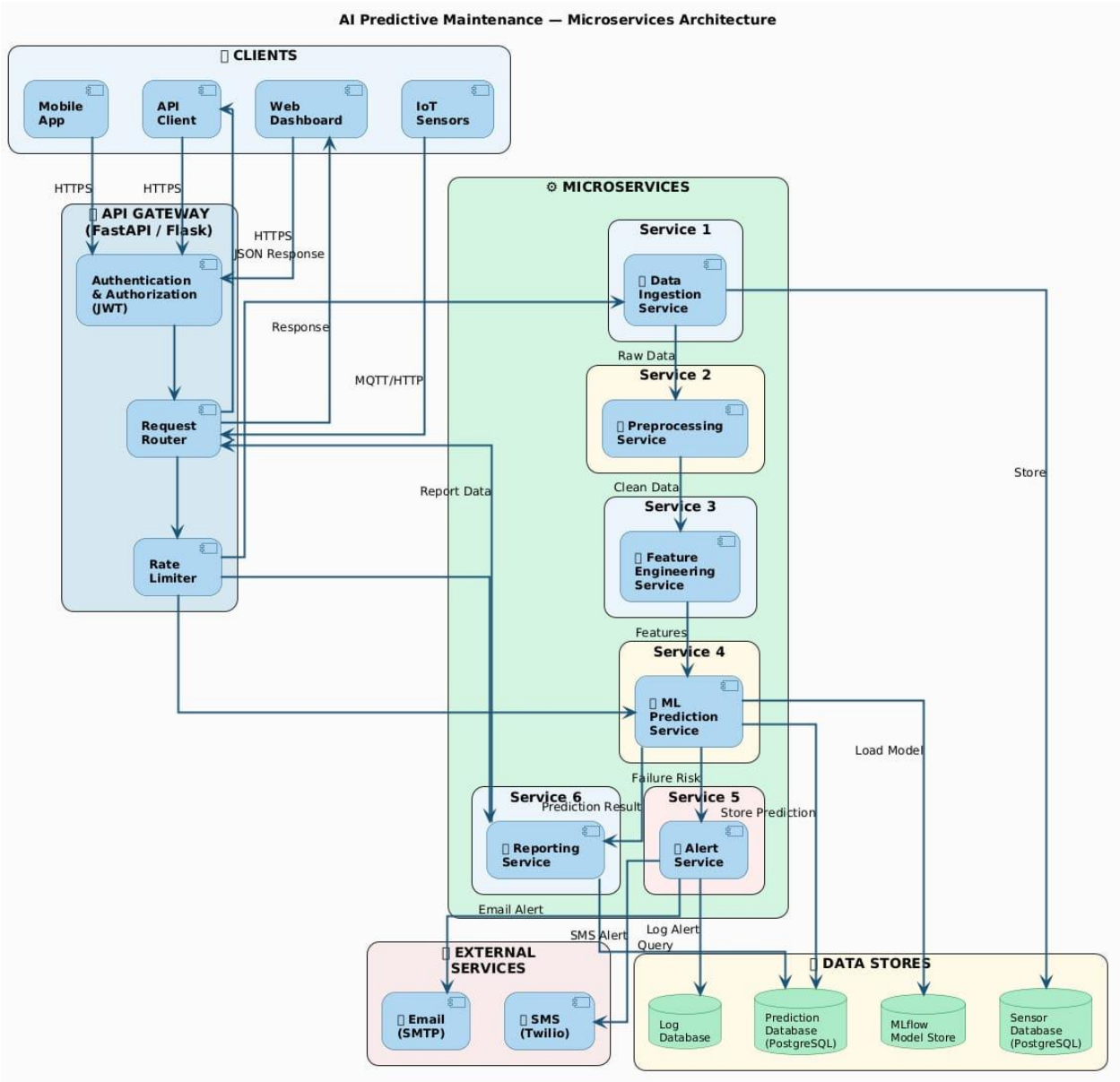
activation diagram



Activity Diagram - Predictive Maintenance Process

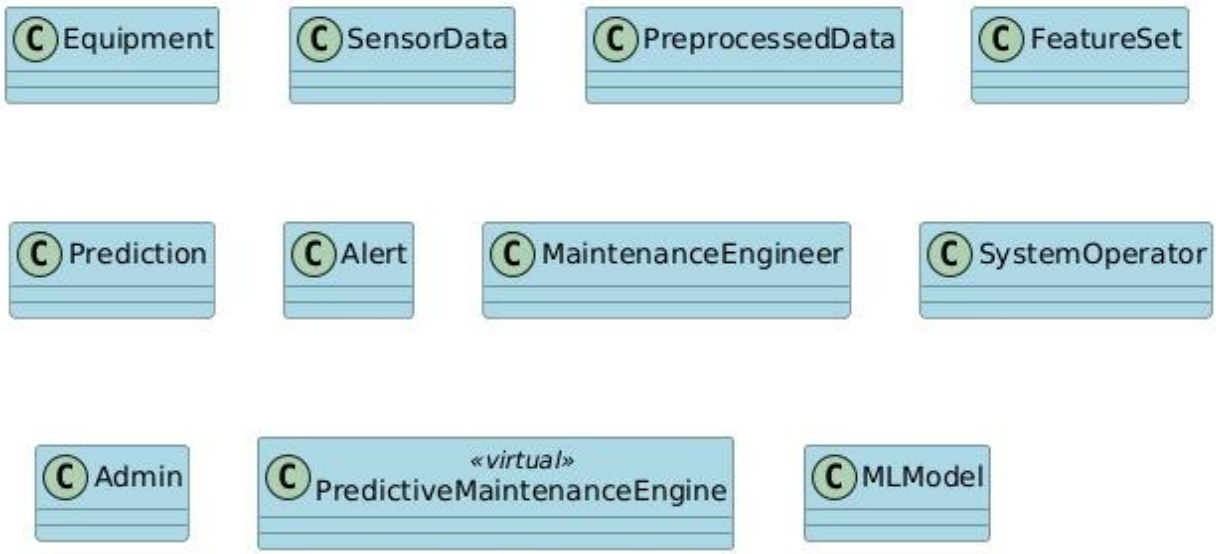


Software diagram



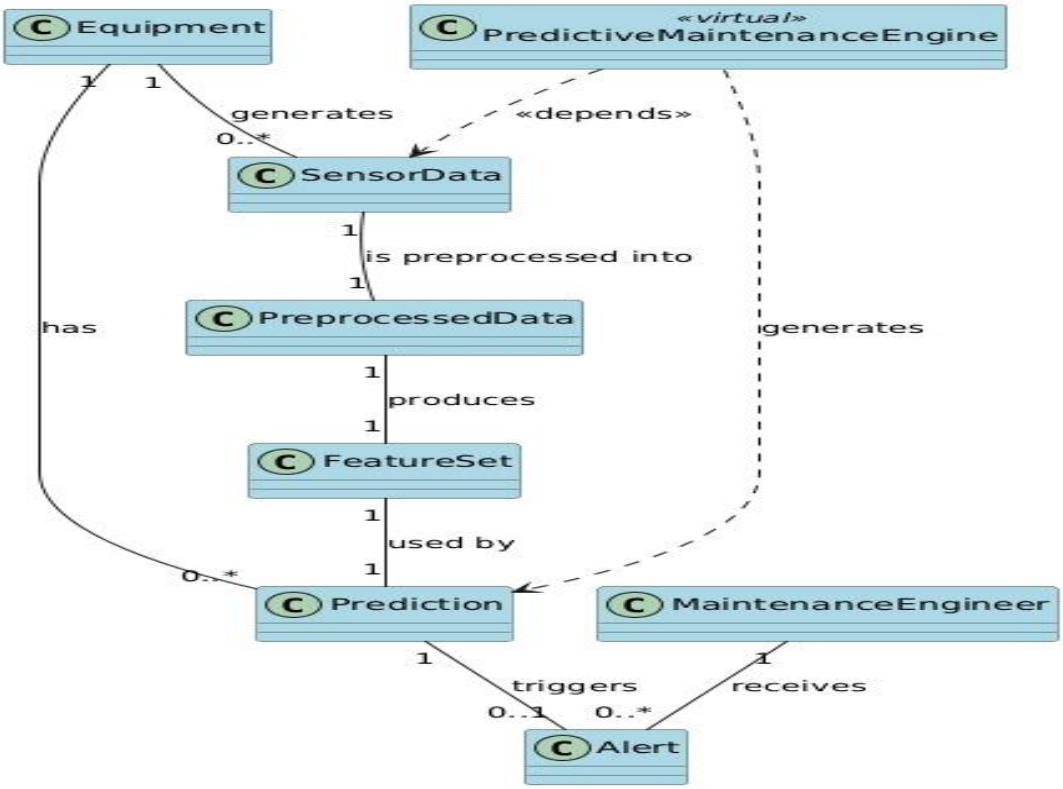
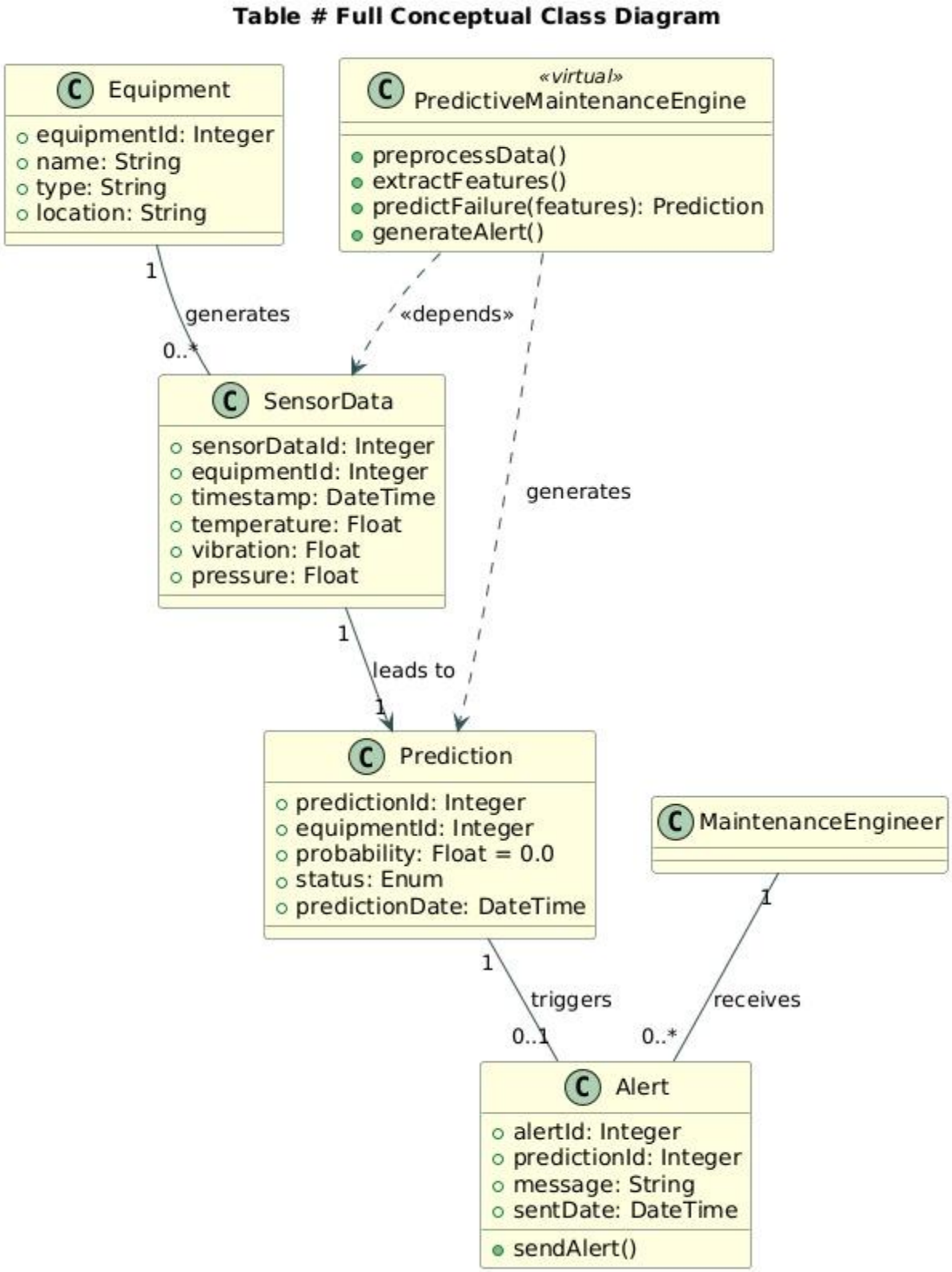
Domain Model

Figure # (Predictive Maintenance System) Conceptual Class Names



Partial Conceptual Class Diagram

Table # Partial Conceptual Class Diagram



Full Conceptual Calss Diagram